



University
of Glasgow

W1-01: What is Big Data?

COMPSCI4064 (H) and COMPSCI5088 (M)

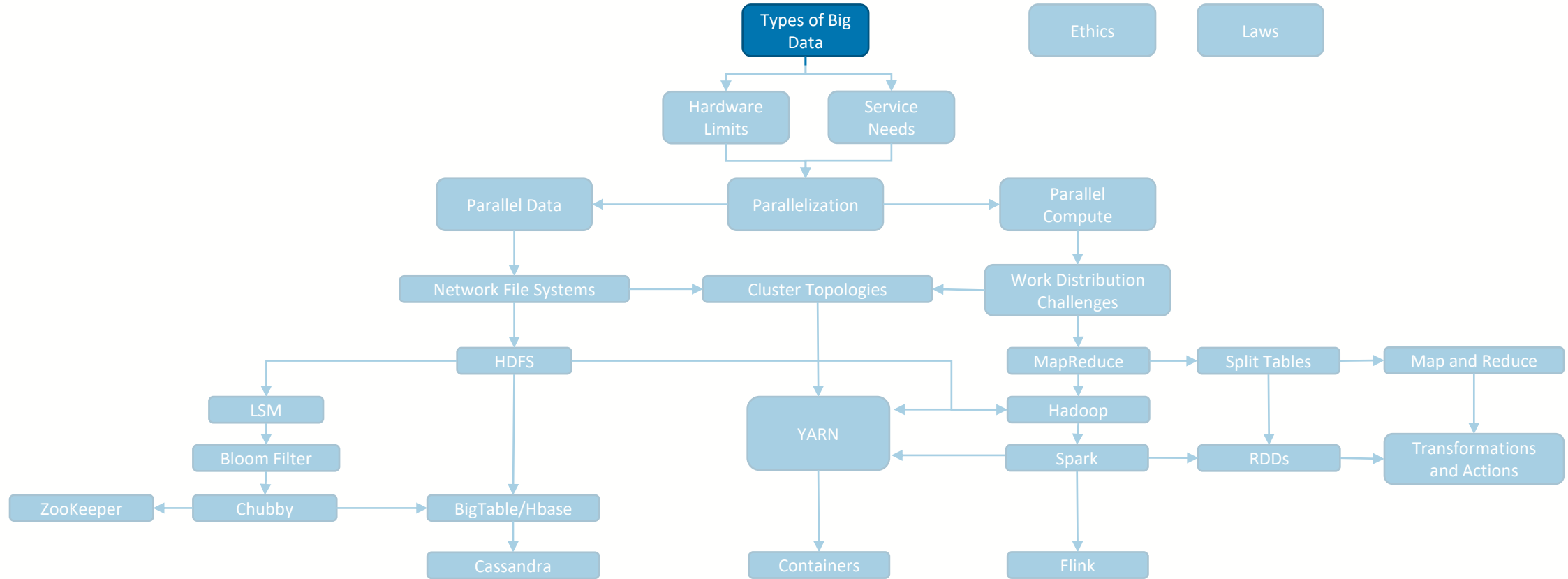
Dr. Richard McCreddie

**WORLD
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Where are we?





What is Big Data?

- “Data whose **capturing, storage, curation, management, and processing** are **not possible** with **traditional tools** within a **tolerable amount of time**”
- Big Data is defined by its *Use*
 - “**capturing, storage, curation, management**” – how we store and access it
 - “**processing**” – how we can use it to gain value
- ... and by the *Software and Hardware* available to us
 - “**traditional tools**” – software
 - “**tolerable amount of time**” – what hardware we have to run that software

APPLICATIONS – ENTERPRISE





Why this explosion of Big Data

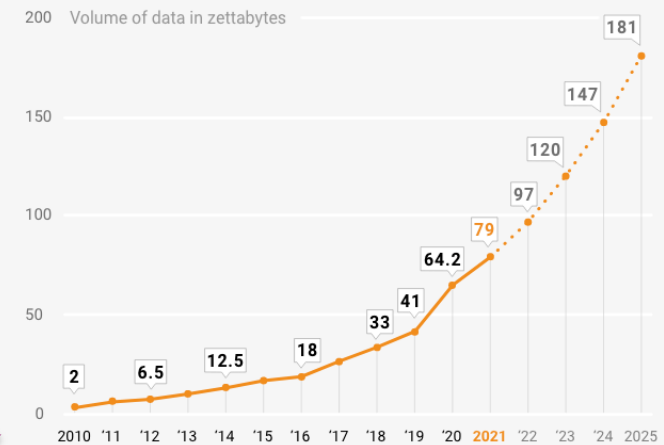
- The simple answer is that we are producing and storing more than we ever have
 - This is a result of a broad acceptance that its better to store data, even if you don't know what you will use it for
 - ... combined with a massive increase in the number and amount of time spent with internet-enabled devices

The up-side of this is there are a LOT of Data Science jobs now

Volume of data created, captured, copied, and consumed worldwide



The volume of data generated, consumed, copied, and stored is projected to exceed 180 zettabytes by 2025



Source: statista.com



firstsiteguide.com



Big Data is Relative

What people might agree on as 'Big Data' changes over time

In the 90's, 100MB might have been considered Big Data

Today 1,000,000,000MB (1 Petabyte) might be Big Data

This shift has happened because our hardware and software has become much more powerful

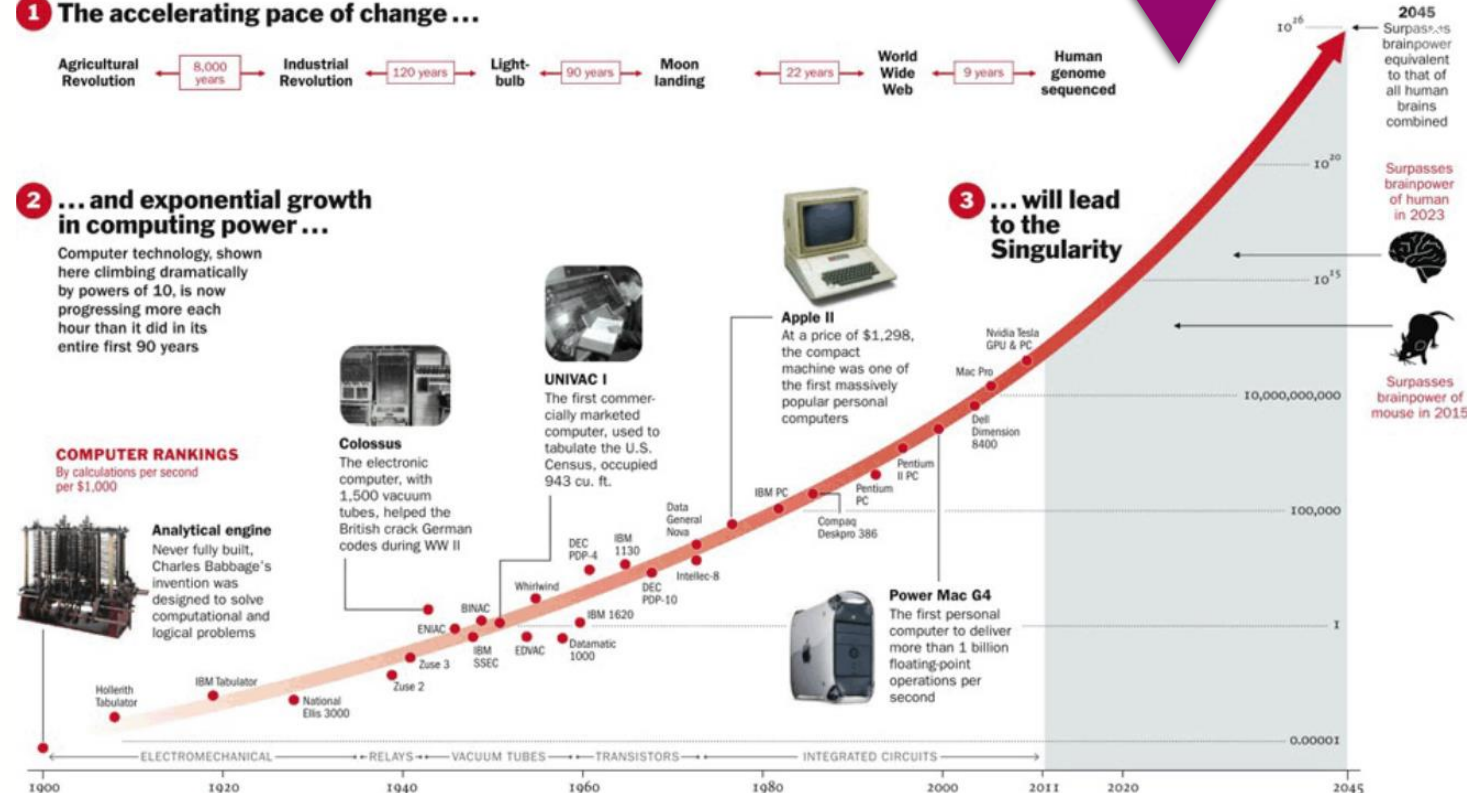
Note that this chart is optimistic... we are approaching the limits of our current technology

1 The accelerating pace of change ...



2 ... and exponential growth in computing power ...

Computer technology, shown here climbing dramatically by powers of 10, is now progressing more each hour than it did in its entire first 90 years



How do you know if you have a Big Data Problem?



Dimensions of Big Data

- **V**olume: (thank you captain obvious...)
- **V**ariety: Structured, semi-structured, unstructured
 - Text, video, audio, photos
 - Wikipedia pages: text + infoboxes
 - Microformat, microdata, (schema.org), ...
- **V**elocity: Near-Real-Time
 - Storage, querying, analytics, ...
- **V**eracity: Is it really the “true” data? errors? alterations?
- **V**ariability: data flows, peaks & valleys, ..
 - Complexity: entities, data, hierarchies, links, relationships, ...

Pick any two and you
are probably working
in a Big Data space



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